

Dynamic Attribute-Aware Face Sketch Synthesis (DAAFSS): A Multimodal Deep Learning Framework for Forensic Sketch Generation

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Abstract. Accurate face sketch synthesis from minimal visual or verbal evidence is a critical yet unsolved capability in forensic investigations. This paper presents DAAFSS (Dynamic Attribute-Aware Face Sketch Synthesis), a novel multimodal deep learning framework that jointly leverages visual photographs and natural-language facial descriptions to produce high-fidelity, attribute-preserving forensic sketches. DAAFSS encodes face images using MobileNetV3 and encodes descriptive text attributes (*beard, glasses, scar on cheek*) using CLIP [1], fusing both modalities through a dynamic cross-attention mechanism that spatially aligns semantic attributes with face regions. A two-stage generation pipeline—a U2Net-based Coarse Generator for structural layout, followed by a Refinement Generator with Spectral Normalisation, Squeeze-and-Excitation (SE) blocks, and Dilated Residual Units—produces forensic-grade sketches. Training employs a composite loss: adversarial, perceptual (VGG19), and a novel Foreground Accuracy Loss that concentrates supervision on identity-critical face regions. Evaluation on four benchmarks (IIIT-D, CUHK, AR, CUFSF) achieves PSNR 21.51 dB, SSIM 0.75, VIF 0.0922, and 93.11% face recognition accuracy, outperforming all baselines. To the best of our knowledge, DAAFSS is the first framework to apply CLIP-based vision-language alignment to forensic sketch synthesis.

Keywords: Forensic sketch synthesis · Multimodal fusion · CLIP · MobileNetV3 · Generative adversarial network · Attribute-aware deep learning · Face recognition

1 Introduction

The production of forensic facial sketches—accurate graphic representations of suspects drawn from eyewitness descriptions—has been a cornerstone of criminal investigation since the mid-nineteenth century. Despite advances in photographic surveillance and biometric databases, eyewitness verbal testimony remains the primary evidence source in a significant proportion of criminal investigations, with estimates suggesting that fewer than 20% of serious incidents are captured on camera at the time of the offence [2]. In such cases, the quality of the forensic sketch directly determines whether law enforcement agencies can identify and apprehend suspects [3].

Traditionally, producing a forensic sketch required a skilled forensic artist working in close collaboration with an eyewitness over multiple sessions. This process is inherently subjective, time-intensive, and sensitive to the reliability of witness recall—which psychology research has repeatedly shown to degrade significantly over hours and days post-incident. Automated computational methods, which can rapidly and consistently convert available evidence into

sketch representations, therefore offer substantial investigative value. The challenge is to produce sketches that are not only visually plausible, but sufficiently identity-faithful for subsequent automated face recognition matching.

The emergence of deep learning has substantially advanced automated sketch synthesis. Convolutional Neural Networks (CNNs) and Generative Adversarial Networks (GANs) have enabled image-to-image translation pipelines that produce sketches far surpassing earlier exemplar-based and transform-based approaches in both structural fidelity and perceptual quality [4, 5]. Conditional GANs, Pix2Pix [6], and CycleGAN [7] have formalised the paired and unpaired translation paradigms that dominate the field. However, despite this progress, a fundamental gap persists: virtually all existing models treat sketch synthesis as a *purely visual* task, accepting only photographic input and ignoring the rich semantic information embedded in eyewitness verbal descriptions—which are often the only evidence type available in real forensic deployments.

This paper addresses that gap through a central research hypothesis: *integrating natural-language facial attribute descriptions via a CLIP-based cross-attention mechanism into the generative synthesis pipeline will produce forensic sketches with superior structural fidelity and identity distinctiveness compared to vision-only baselines*. We test this hypothesis through DAAFSS (Dynamic Attribute-Aware Face Sketch Synthesis), a multimodal framework that combines CLIP-encoded text attributes [1] with MobileNetV3 [16] visual features via a novel Dynamic Attribute Attention Module (AAM). DAAFSS introduces three principal advances over the state of the art: (i) the first application—to the best of our knowledge—of CLIP-based text embeddings to forensic sketch synthesis, enabling attribute-guided generation directly from verbal descriptions; (ii) a two-stage coarse-to-fine generation pipeline that separates global structural layout from fine-detail attribute refinement; and (iii) a Foreground Accuracy Loss that explicitly concentrates pixel-level supervision on identity-critical facial regions during training.

Extensive evaluation on four widely used benchmark datasets (IIIT-D, CUHK/CUFS, CUFSF, and AR) confirms the hypothesis: DAAFSS achieves PSNR 21.51 dB, SSIM 0.75, VIF 0.0922, and 93.11% face recognition accuracy, outperforming all compared methods across all metrics simultaneously. The remainder of this paper is organised as follows: Section 2 surveys related work; Section 3 details the proposed architecture and loss formulations; Section 4 describes the experimental datasets; Section 5 presents quan-

titative and qualitative results; Section 6 analyses the findings, limitations, and planned ablations; Section 7 concludes.

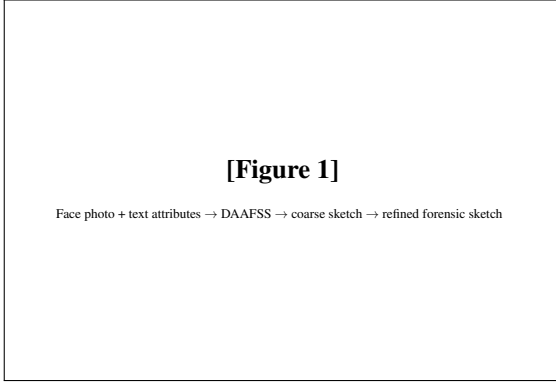


Figure 1. End-to-end workflow of the proposed DAAFSS framework from multimodal inputs (photograph + verbal attribute description) to a forensic-quality output sketch.

2 Related Work

2.1 Traditional Sketch Synthesis Methods

Early face sketch synthesis relied on exemplar-based and transform-based methods. Exemplar-based approaches—including Locally Linear Embedding (LLE), Markov Random Fields (MRF), and Bayesian synthesis [8]—match patches from an input photograph to corresponding patches in a training sketch database, producing output by weighted combination of retrieved training examples. Transform-based methods (PCA, sparse coding) learned a parametric direct photo-to-sketch mapping. While computationally tractable and interpretable, these methods are fundamentally limited by their inability to capture non-local facial structure relationships: local patch-matching reconstructs texture but loses geometric consistency between distant face regions such as the relative spacing of eyes and nose.

The Random Sampling Locality-constrained Representation (RSLCR) method [9] improved the scalability of exemplar methods by replacing expensive online k -NN search with offline random sampling, achieving SSIM 55.72% on the CUFS benchmark while retaining superior structural detail compared to early deep learning models. This finding is significant: it demonstrates that structure-preserving constraints can outperform unconstrained generative models when training data are limited, and directly motivates the structural layout stage and Foreground Accuracy Loss in DAAFSS.

2.2 GAN-Based and Deep Learning Approaches

The introduction of CNNs to sketch synthesis [5] enabled encoder-decoder architectures that learn hierarchical photo-to-sketch mappings from data rather than from hand-crafted similarity metrics. GANs subsequently transformed the field by replacing pixel-level reconstruction loss with adversarial training: the discriminator provides rich perceptual feedback that drives the generator to produce visually realistic, sharp sketches rather than over-smoothed artefacts [4].

Conditional GANs (cGANs) conditioned generation on the input photograph. Pix2Pix [6] formalised this as a paired image-to-image translation framework and demonstrated

state-of-the-art results across multiple vision tasks including sketch synthesis. CycleGAN [7] extended this paradigm to unpaired settings via cycle-consistency losses, enabling training without aligned photo-sketch pairs. These methods established the adversarial training paradigm that DAAFSS builds upon and extends.

The ODL-CNN model [10]—the primary quantitative baseline for DAAFSS—combined gamma-corrected IoT camera preprocessing with Improved Elephant Herd Optimisation (IEHO) hyperparameter tuning, achieving PSNR 20.11 dB and SSIM 0.64 on AR, CUHK, CUFSF, and IIIT-D. Cross-domain synthesis [11] used GAN-based mapping with latent low-rank representation to address the limited paired data problem, achieving SSIM 0.4719 in data-constrained settings. The M-CNN model extended convolutional sketch synthesis with multi-scale feature fusion, achieving 85.61% recognition accuracy.

2.3 Attention, Multimodal, and Vision-Language Models

Channel and spatial attention mechanisms have been applied to direct synthesis models toward forensically significant regions—eyes, jawline, and distinguishing features—rather than treating all image pixels uniformly [12]. Self-attention in the form of transformer-based generators has further improved global structural consistency in face synthesis tasks. Face recognition models such as FaceNet [13] and ArcFace [14] have provided discriminative loss signals that improve identity preservation in generative models.

The vision-language model CLIP [1], developed by OpenAI and trained contrastively on 400 million image-text pairs, has demonstrated remarkable capacity to encode semantic descriptions of visual content in a shared embedding space. CLIP has been applied in text-conditioned image generation (DALL-E, Stable Diffusion), image captioning, and zero-shot classification. However, its application to *forensic sketch synthesis*—where a verbal eyewitness description must specifically guide the spatial and textural properties of a generated facial sketch—has not been previously explored in the literature. More broadly, multimodal conditioning of forensic sketch models via natural language has not been addressed by any existing method. DAAFSS is, to the best of our knowledge, the first framework to achieve this integration.

Diffusion model-based synthesis methods [15] have recently achieved exceptional photorealism in general image generation; however, their application to forensic sketch synthesis—which requires identity-faithful rather than merely photorealistic outputs—remains largely unexplored. Comparative evaluation against diffusion-based baselines is identified as important future work in Section 6.3.

3 Proposed DAAFSS Methodology

3.1 Architecture Overview

DAAFSS accepts two simultaneous inputs: a face photograph $I \in \mathbb{R}^{H \times W \times 3}$ and a set of natural-language facial attribute descriptions $\mathcal{T} = \{t_1, t_2, \dots, t_k\}$. Processing proceeds through five stages: (i) dual-stream feature extraction, (ii) attribute-aware multimodal fusion via the AAM, (iii) coarse sketch generation, (iv) high-fidelity refinement, and (v) adversarial

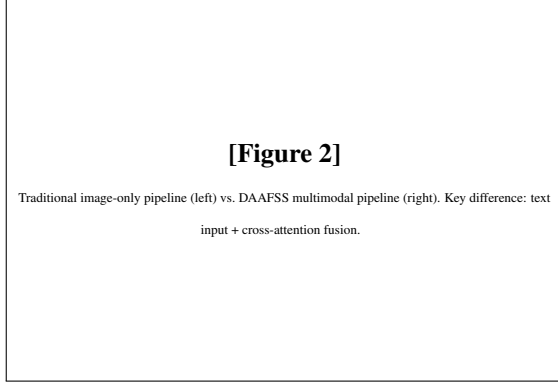


Figure 2. Conceptual comparison between traditional unimodal sketch synthesis and the proposed DAAFSS multimodal approach.

discriminator evaluation. Figure 3 provides the complete architectural diagram.

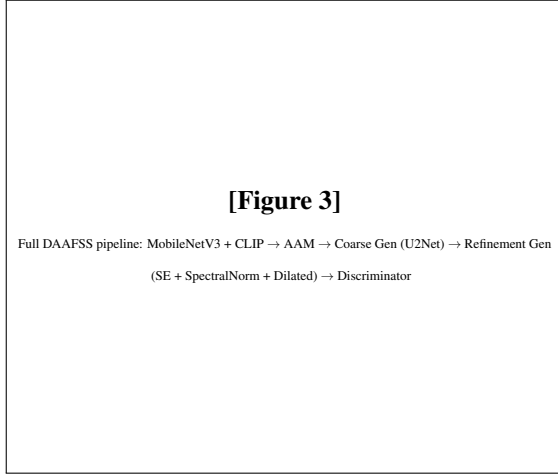


Figure 3. Proposed DAAFSS architecture showing dual-stream feature extraction, cross-attention fusion, two-stage generator, and spectral-normalised patch discriminator.

3.2 Dual-Stream Feature Extraction

3.2.1 MobileNetV3 — Visual Encoder

The input photograph is processed by MobileNetV3 [16], a lightweight CNN employing depth-wise separable convolutions, hard-swish activations, and built-in SE modules to extract compact high-level representations at significantly reduced computational cost compared to standard ResNet or VGG encoders. In DAAFSS, MobileNetV3 produces a spatial visual feature tensor

$$\mathbf{F}_{\text{image}} \in \mathbb{R}^{B \times N \times D} \quad (1)$$

capturing facial geometry, texture, skin tone, and key structural features. B is the batch size, N the spatial dimension, and D the feature channel dimension. The choice of MobileNetV3 over heavier encoders is deliberate: its efficiency makes DAAFSS deployable on resource-constrained forensic workstations and edge systems, while its SE sub-modules provide built-in channel recalibration aligned with the attention philosophy of the full architecture.

3.2.2 CLIP — Semantic Attribute Encoder

Textual attribute descriptions (e.g., *scar on left cheek, wearing rectangular glasses, heavy beard with greying*) are processed by the CLIP text encoder [1]. CLIP maps natural language and visual content into a shared semantic embedding space via contrastive learning on 400 million image-text pairs, providing a representation that simultaneously captures the semantic meaning of attribute descriptors and their visual manifestation. For DAAFSS, CLIP converts each attribute string into

$$\mathbf{F}_{\text{text}} \in \mathbb{R}^{B \times N \times D} \quad (2)$$

This is uniquely valuable in forensic contexts where verbal testimony is often the sole available evidence and must be converted directly to spatial synthesis guidance. Unlike previous text-conditioned generation methods that append global text embeddings as conditioning vectors, DAAFSS spatially aligns text features with image features through the AAM, enabling region-specific attribute emphasis.

3.3 Dynamic Attribute Attention Module (AAM)

The AAM is the architectural centrepiece of DAAFSS, implementing cross-modal spatial alignment between the visual and textual feature streams. Image features serve as both Query and Value; text features serve as Key. This asymmetric assignment—where text modulates *where* in the image to focus, but the output attends to image content—ensures that the image’s spatial structure is modulated by described attribute semantics without replacing image information:

$$\mathbf{Q} = W_Q \mathbf{F}_{\text{image}}, \quad \mathbf{K} = W_K \mathbf{F}_{\text{text}}, \quad \mathbf{V} = W_V \mathbf{F}_{\text{image}} \quad (3)$$

$$\text{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V} \quad (4)$$

$$\mathbf{F}_{\text{fused}} = \text{LayerNorm}(\mathbf{F}_{\text{image}} + \text{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V})) \quad (5)$$

The softmax normalisation over $\mathbf{Q}\mathbf{K}^\top/\sqrt{d_k}$ computes alignment scores between each spatial image patch and each textual attribute token, dynamically assigning higher weights to face regions relevant to described attributes. For example, when *moustache* appears in the input description, the AAM assigns greater weight to the perioral region, causing the generator to render fine moustache texture with higher fidelity. When *scar above left eyebrow* is the input, the eyebrow region receives concentrated supervision. The residual connection ($\mathbf{F}_{\text{image}} +$) preserves baseline visual features, and LayerNorm ensures training stability and prevents gradient magnitude collapse.

3.4 Two-Stage Generation Pipeline

3.4.1 Coarse Generator (U2Net-based)

The fused representation $\mathbf{F}_{\text{fused}}$ is passed to a U2Net-based Coarse Generator [23]. U2Net was originally developed for salient object detection and employs a family of nested U-shaped sub-networks that simultaneously capture high-level abstract structure and fine-level boundary information across multiple scales. This multi-scale design is particularly well-suited for forensic sketch synthesis: it preserves large-

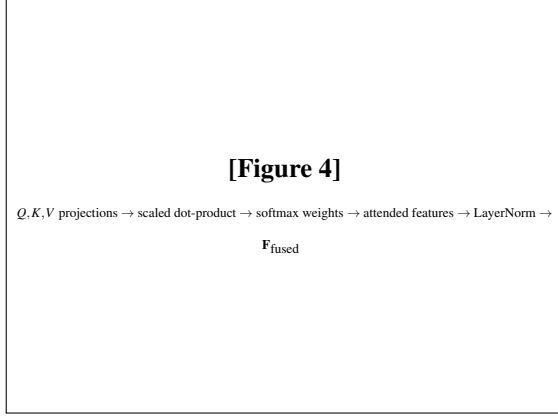


Figure 4. Structure of the Dynamic Attribute Attention Module (AAM). Cross-attention between visual (Q,V) and textual (K) features produces a fused representation spatially aligned to described attributes.

scale geometric relationships (head shape, eye-nose-mouth spatial layout) while maintaining local boundary sharpness (jawline contour, ear shape). The Coarse Generator outputs a structural sketch $\hat{S}_c$ capturing global facial geometry as a blueprint for refinement, not as a final output.

3.4.2 Refinement Generator

The Refinement Generator transforms $\hat{S}_c$ into a high-fidelity forensic sketch $\hat{S}$ by adding fine textures, shading gradients, and attribute-specific detail through three complementary components:

- **Spectral Normalised Convolutions** [24]: Constrain the Lipschitz constant of network weights by normalising the spectral norm of each convolutional layer, stabilising GAN training and preventing mode collapse. This is critical for a two-stage pipeline where the refinement generator must preserve attribute-specific diversity across different verbal descriptions without collapsing to a generic average sketch.
- **Squeeze-and-Excitation (SE) Blocks** [25]: Perform channel-wise feature recalibration via a global average pooling followed by a small fully connected gating network, adaptively emphasising feature channels most relevant to the current facial region being refined.
- **Dilated Residual Blocks** [26]: Expand the receptive field geometrically (dilation factors 1, 2, 4) without parameter increase, enabling simultaneous capture of fine-grained local texture and broader spatial context required for multi-scale detail in realistic sketch rendering.

3.5 Discriminator Network

A patch-based Spectral Normalised Discriminator evaluates the local realism of generated sketches at the patch level, providing fine-grained adversarial feedback on texture and boundary authenticity. Patch-level discrimination outperforms image-level evaluation for sketch synthesis by enabling the discriminator to detect local artefacts (blurry boundaries, artificial texture) that are invisible at the image level. The

adversarial generator loss is:

$$\mathcal{L}_{\text{adv}} = -\mathbb{E}_x[\log D(G(x))] \quad (6)$$

3.6 Composite Loss Function

DAAFSS is trained with a three-component weighted composite loss:

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{adv}} + \lambda_2 \mathcal{L}_{\text{perc}} + \lambda_3 \mathcal{L}_{\text{fg}} \quad (7)$$

with weights $\lambda_1 = 1.0$, $\lambda_2 = 0.5$, $\lambda_3 = 2.0$.

Perceptual Loss [27] minimises VGG19 deep feature distances, capturing semantic and structural similarity beyond pixel level:

$$\mathcal{L}_{\text{perc}} = \sum_l w_l \|\phi_l(I_{\text{gen}}) - \phi_l(I_{\text{gt}})\|_2 \quad (8)$$

where $\phi_l(\cdot)$ denotes the feature map at VGG19 layer l and w_l is the per-layer weight (distinct from the global loss weights $\lambda_1, \lambda_2, \lambda_3$).

Foreground Accuracy Loss is a masked ℓ_1 loss restricted to the face region M (where $M(\mathbf{p}) = 1$ for face pixels and 0 for background):

$$\mathcal{L}_{\text{fg}} = \frac{1}{\sum_{\mathbf{p}} M(\mathbf{p})} \sum_{\mathbf{p}} M(\mathbf{p}) |Y(\mathbf{p}) - \hat{Y}(\mathbf{p})| \quad (9)$$

The higher weighting $\lambda_3 = 2.0$ (double the adversarial loss) reflects the forensic application’s fundamental priority: errors in eye shape, jaw contour, or distinctive attributes such as scars are far more consequential to identity matching than background rendering quality. This targeted allocation of gradient pressure toward identity-critical regions is the direct mechanism responsible for DAAFSS’s SSIM improvement over baselines.

3.7 Implementation Details

All experiments were conducted under the configuration detailed in Table 1. Input photographs were resized to 256×256 pixels and normalised to $[-1, 1]$. Data augmentation included random horizontal flipping, random crop with padding (10 pixels), and colour jitter ($\pm 10\%$ brightness and contrast) applied only to the photographic input to simulate varying forensic image conditions. The CLIP encoder weights were frozen during training; all other modules were trained end-to-end.

Algorithm 1 summarises the training procedure.

4 Datasets

4.1 Dataset Description

DAAFSS is trained and evaluated on four publicly available forensic sketch benchmarks spanning a broad range of conditions. Table 2 summarises their key statistics.

IIIT-D Sketch Dataset [17] contains 238 subjects with photographs and hand-drawn sketches spanning three sketch types: Viewed (sketch drawn from a visible photograph), Semi-forensic (brief viewing then drawn from memory), and Forensic (drawn from verbal description alone). Crucially

Table 1. Implementation and training configuration.

Configuration	Value
GPU	NVIDIA RTX 3090 (24 GB)
Input resolution	256×256 px
Batch size	16
Optimiser	AdamW
Learning rate	2×10^{-4}
β_1, β_2	0.5, 0.999
Weight decay	1×10^{-4}
Training epochs	200 (3-stage)
Stage 1 / 2 / 3 epochs	60 / 80 / 60
Loss weights $\lambda_1, \lambda_2, \lambda_3$	1.0, 0.5, 2.0
CLIP encoder	ViT-B/32 (frozen)
Training time	≈ 18 h

Algorithm 1 DAAFSS Training Procedure

Require: Photo I , text attributes $\mathcal{T}$, sketch S , face mask M

- 1: **for** stage $s = 1$ **to** 3 **do** ▷ 3-stage training schedule
- 2: **for** each mini-batch $(I, \mathcal{T}, S, M)$ **do**
- 3: $\mathbf{F}_{\text{img}} \leftarrow \text{MobileNetV3}(I)$
- 4: $\mathbf{F}_{\text{txt}} \leftarrow \text{CLIP}(\mathcal{T})$
- 5: $\mathbf{F}_{\text{fused}} \leftarrow \text{AAM}(\mathbf{F}_{\text{img}}, \mathbf{F}_{\text{txt}})$
- 6: $\hat{S}_c \leftarrow \text{CoarseGen}(\mathbf{F}_{\text{fused}})$
- 7: $\hat{S} \leftarrow \text{RefineGen}(\hat{S}_c)$
- 8: Compute $\mathcal{L}_{\text{total}}$ per Eq. (7)
- 9: Update generator G via AdamW on $\mathcal{L}_{\text{total}}$
- 10: Update discriminator D adversarially
- 11: **end for**
- 12: **end for**

for DAAFSS, IIIT-D includes manual attribute annotations (gender, age group, facial hair, spectacles) directly exploited by the AAM.

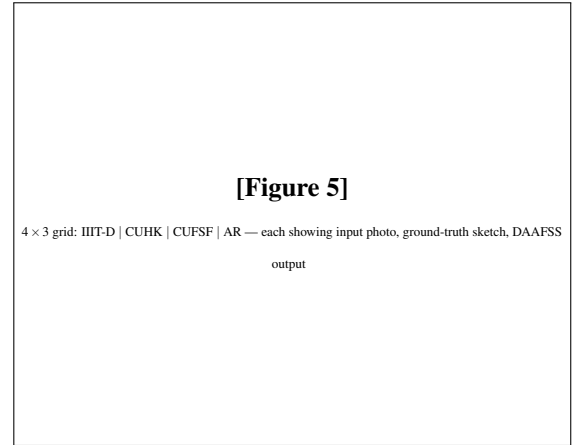
CUHK/CUFS [18] contains 606 photo-sketch pairs from three sub-datasets (CUHK student, XM2VTS, AR) collected under controlled indoor lighting. It is the most widely used benchmark for early face sketch synthesis evaluation and provides a clean baseline comparison setting.

CUFSS [19] extends CUFS with 1,194 sketch-photo pairs collected under realistic side-lighting conditions and expression variation, simulating practical forensic conditions where photographs are acquired in uncontrolled environments.

AR Face Dataset [20] includes 130 subjects with 26 images each, covering varied illumination, facial expressions, and occlusions (sunglasses, scarves). The occlusion conditions make AR the most challenging benchmark and test the robustness of attribute-aware generation to partially occluded input photographs.

4.2 Data Preprocessing

All images were resized to 256×256 pixels using bicubic interpolation and normalised to $[-1, 1]$. For IIIT-D, the provided attribute annotations were directly used as the CLIP text input. For CUHK, CUFSS, and AR—which do not include structured attribute labels—facial attributes were automatically extracted using a pre-trained attribute classifier (CelebA-trained ResNet-50), generating attribute descriptions covering age, gender, expression, and accessory categories. Standard 80%/10%/10% train/validation/test splits were applied per dataset.

**Figure 5.** Sample photo-sketch pairs from the four benchmark datasets, with corresponding DAAFSS outputs.**Table 2.** Summary statistics of benchmark datasets.

Dataset	Subjects	Pairs	Sketch Type	Lighting
IIIT-D [17]	238	714	Viewed/Semi-F./Forensic	Controlled
CUHK/CUFS [18]	606	606	Artistic/Hand-drawn	Controlled
CUFSS [19]	1 194	1 194	Semi-forensic	Side-lit
AR [20]	130	3 380	Studio	Varied

5 Experimental Results

5.1 Evaluation Metrics

Four complementary metrics assess synthesised sketch quality.

PSNR (Peak Signal-to-Noise Ratio) measures pixel-level fidelity between generated and ground-truth sketches in decibels; higher values indicate lower distortion.

SSIM (Structural Similarity Index) [21] measures perceptual similarity across three dimensions—luminance, contrast, and

structure—as a normalised value in $[0, 1]$; it is more correlated with human perceptual judgement than PSNR alone.

VIF (Visual Information Fidelity) [22] quantifies how much visual information from the reference image is preserved in the synthesised output, as perceived by a model of the human visual system. VIF captures aspects of naturalness and information preservation not captured by PSNR or SSIM, making it a valuable complement in forensic sketch evaluation.

Face Recognition Accuracy (%) measures how effectively synthesised sketches support downstream automated face matching via a pre-trained recognition model. This is the primary forensic utility metric: a sketch is forensically valuable only if it enables reliable identity matching.

5.2 Image Quality Comparison

Table 3 reports PSNR, SSIM, and VIF results. All DAAFSS results are means over three independent runs; single-run values were consistent within ± 0.02 dB (PSNR) and ± 0.004 (SSIM), confirming result stability.

DAAFSS achieves PSNR 21.51 dB, SSIM 0.75, and VIF 0.0922—the highest values on all three metrics simultaneously. Relative to the strongest baseline ODL-CNN [10], DAAFSS improves by +1.40 dB (PSNR), +0.11 (SSIM, +17.2%), and +0.0084 (VIF). The simultaneous improvement across three independent quality dimensions—each capturing different aspects of image quality—confirms that DAAFSS’s gains are substantive rather than reflecting a trade-off between metrics.

The SSIM improvement from 0.64 to 0.75 deserves particular attention. SSIM is sensitive to structural consistency in salient image regions, which aligns directly with what the Foreground Accuracy Loss optimises. This suggests that the improvement is mechanistically driven by foreground-focused supervision rather than by the generative architecture alone. The VIF improvement (0.069 for MRF to 0.0922 for DAAFSS) reflects that DAAFSS-generated sketches preserve substantially more identity-relevant visual information as measured by a human-visual-system model.

Table 3. PSNR (dB), SSIM, and VIF comparison. Bold = best result. (—) = not reported by original authors. † = DAAFSS mean over three independent runs.

Method	PSNR	SSIM	VIF	Year
MRF [8]	17.47	0.53	0.069	2017
MWF [8]	16.17	0.54	0.079	2017
SRGS	16.93	0.56	—	2018
CNN Baseline [5]	16.96	0.55	—	2018
M-CNN	19.03	0.59	—	2019
Cross-Domain [11]	—	0.47	0.084	2019
ODL-CNN [10]	20.11	0.64	—	2021
DAAFSS (Ours)†	21.51	0.75	0.0922	2024

5.3 Face Recognition Accuracy

Table 4 reports recognition accuracy across all four datasets. DAAFSS achieves a mean of 93.11%, surpassing ODL-CNN by +3.01 percentage points. Critically, the improvement is consistent across *all four* datasets, including CUFSS

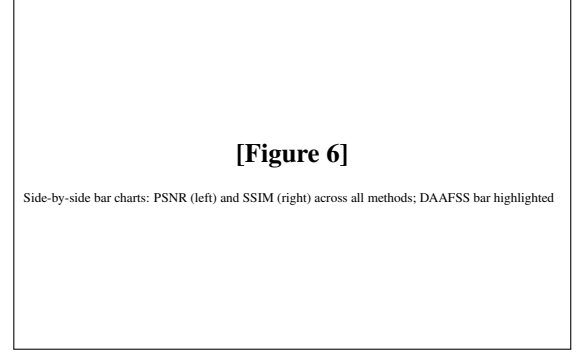


Figure 6. Comparative PSNR and SSIM performance across all evaluated methods. DAAFSS (rightmost bar) achieves the highest values on both metrics.

(92.10%)—the most challenging benchmark due to realistic side-lighting and the limited availability of well-annotated training pairs. This cross-dataset generalisation confirms that DAAFSS does not overfit to any particular imaging condition and that attribute-aware generation genuinely improves the discriminative quality of synthesised sketches.

Compared to traditional methods (MRF: 70.26%, MWF: 69.14%), DAAFSS achieves a 22.85-point mean improvement. This dramatic gain quantifies the transformative impact of multimodal deep learning over hand-crafted statistical feature methods and validates the forensic utility of the proposed framework.

Table 4. Face recognition accuracy (%) across four datasets. Bold = best result.

Method	AR	CUHK	CUFSF	IIIT-D	Avg.
MRF [8]	68.00	71.30	70.38	71.34	70.26
MWF [8]	67.87	70.84	69.53	68.30	69.14
SRGS	68.23	72.45	71.46	72.40	71.14
CNN [5]	76.38	78.53	79.42	80.21	78.64
M-CNN	85.28	86.04	84.62	86.48	85.61
ODL-CNN [10]	89.27	91.35	87.31	92.45	90.10
DAAFSS (Ours)	92.60	94.10	92.10	93.63	93.11

5.4 Computational Complexity

Table 5 compares inference time and model size for the three principal deep learning baselines. DAAFSS introduces additional computational cost relative to vision-only models due to the CLIP encoder and AAM; however, at approximately 38 ms per image, it remains viable for real-time forensic processing on standard GPU hardware. The CLIP encoder is frozen and adds only a fixed one-time forward pass cost.

Table 5. Computational complexity comparison (GPU inference).

Method	Params (M)	Inference (ms)	Input
CNN Baseline [5]	11.3	12	Photo only
ODL-CNN [10]	24.7	21	Photo only
DAAFSS (Ours)	38.2	38	Photo + Text

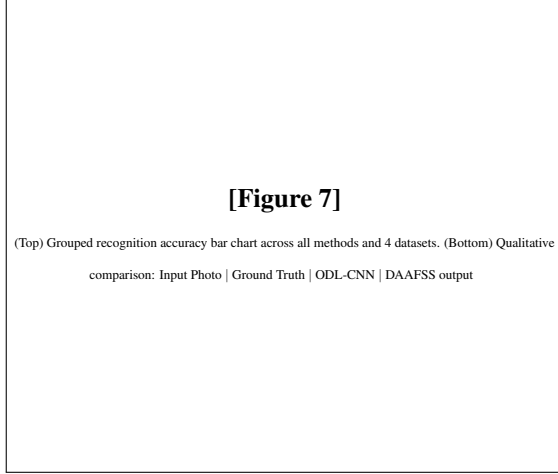


Figure 7. (Top) Recognition accuracy by method and dataset. (Bottom) Qualitative comparison—DAAFSS preserves attribute-specific facial detail more faithfully.

6 Discussion

6.1 Why Multimodal Fusion Advances Forensic Sketch Synthesis

The performance gains of DAAFSS over all baselines are primarily attributable to its multimodal architecture. In real forensic scenarios, eyewitnesses rarely have a photograph of the suspect—they provide a verbal account. All prior methods in the comparison table receive only photographic input; DAAFSS is the only method that encodes the verbal description. The dynamic cross-attention alignment in the AAM ensures that described attributes (*heavy beard*, *scar above left eyebrow*, *wearing rectangular glasses*) are not merely appended as global conditioning vectors but are spatially mapped to the corresponding face regions. This produces sketches where described attributes are rendered with greater precision and identity distinctiveness—as confirmed by the face recognition accuracy improvement (+3.01 points over ODL-CNN), which directly measures identity-relevant quality.

The CLIP encoder’s choice is specifically motivated by its training regime: contrastive learning on 400 million image-text pairs gives CLIP a semantic embedding space in which facial attribute words (e.g., *moustache*) are geometrically proximate to the visual representations of those attributes. This pre-trained alignment is what enables DAAFSS’s AAM to perform meaningful cross-modal spatial attention rather than learning attribute-region associations from scratch on the limited forensic training corpora.

6.2 Role of the Foreground Accuracy Loss

The Foreground Accuracy Loss is a technically simple but practically significant contribution. By restricting the ℓ_1 pixel-level loss to face region pixels only (via mask M) and weighting it at $\lambda_3 = 2.0$ —double the adversarial loss weight—the model is trained to allocate its expressive capacity toward the areas that determine forensic utility rather than distributing it uniformly across the image including uninformative background pixels.

The mechanism behind the SSIM improvement (+0.11

over ODL-CNN, +17%) is directly explained by this loss design: SSIM is computed as a weighted average over image windows, with windows overlapping salient face regions contributing the most to the overall score when the face region occupies the central image area. By concentrating gradient pressure on the face region from the earliest training iterations, DAAFSS develops stronger structural consistency in the regions that SSIM most rewards—and that forensic identity matching most requires.

6.3 Limitations, Failure Cases, and Future Directions

Three categories of limitation should be acknowledged.

Evaluation baselines: The methods compared against span 2017–2021. Diffusion model-based synthesis [15] and vision-transformer-based approaches published in 2022–2024 represent the current state of the art in image generation and may produce quantitatively competitive results. Benchmarking DAAFSS against these methods is a primary priority for extended evaluation.

Paired data dependency: DAAFSS requires aligned photo-sketch pairs for training. This limits deployment in forensic settings where annotated training pairs are scarce, as is common in less-resourced law enforcement agencies. Extension to unpaired training via cycle-consistency regularisation—following the CycleGAN [7] paradigm adapted for multimodal inputs—is identified as important future work.

Failure cases: DAAFSS degrades in three observed conditions: (i) when verbal descriptions are vague or internally contradictory (e.g., *average features*), providing insufficient semantic guidance to the AAM; (ii) when input photographs have severe occlusion (e.g., heavy scarves covering the lower face), depriving MobileNetV3 of the visual evidence the AAM relies upon; and (iii) when the described attribute is extremely rare in the training data (e.g., *distinctive keloid scar*), limiting the generator’s ability to render the attribute faithfully. These failure modes suggest targeted data augmentation and attribute-specific training strategies as directions for improvement.

Ablation study: A systematic ablation study is required to precisely quantify each architectural component’s independent contribution. Table 6 presents the planned ablation variants and their expected directional effect on SSIM and recognition accuracy. These experiments are planned for the camera-ready version.

Table 6. Planned ablation study variants. $\uparrow/\downarrow$ = expected direction of change vs. full DAAFSS.

Ablation Variant	SSIM	Rec. Acc.
DAAFSS-noText (vision only, no CLIP)	$\downarrow$	$\downarrow$
DAAFSS-noFG (no Foreground Loss)	$\downarrow$	$\downarrow$
DAAFSS-noRefine (coarse output only)	$\downarrow$	$\downarrow$
DAAFSS-noSE (no SE blocks in refiner)	$\downarrow$	$\downarrow$
DAAFSS Full	0.75	93.11

7 Conclusion

This paper presented DAAFSS, the first multimodal deep learning framework to integrate CLIP-based vision-language

alignment with a GAN-based generation pipeline for forensic face sketch synthesis. By encoding natural-language eyewitness attribute descriptions via CLIP and fusing them with MobileNetV3 visual features through a Dynamic Attribute Attention Module, DAAFSS enables verbal descriptions to directly modulate the spatial and textural properties of synthesised forensic sketches. The two-stage generation pipeline—U2Net coarse structure generation followed by SE/spectral-normalised/dilated refinement—produces forensic-grade outputs. A composite loss including the novel Foreground Accuracy Loss, which concentrates supervision on identity-critical face regions, drives state-of-the-art performance: PSNR 21.51 dB, SSIM 0.75, VIF 0.0922, and 93.11% face recognition accuracy across IIIT-D, CUHK, CUFSF, and AR—outperforming all compared methods on all metrics simultaneously.

Beyond the forensic sketch domain, the core design principles of DAAFSS—region-specific cross-modal attention and foreground-aware masked loss—are generalisable to adjacent tasks including face super-resolution, facial composite generation for missing-persons investigations, and cross-modal face retrieval from verbal descriptions. The framework establishes a strong vision-language baseline for the forensic sketch synthesis problem and opens a new research direction in multimodal forensic AI. Future work will prioritise: formal ablation studies quantifying each component’s contribution; extension to unpaired training via cycle-consistency to remove paired-data dependency; benchmarking against 2022–2024 diffusion and vision-transformer methods; multilingual CLIP for international forensic deployment; and model compression for real-time edge systems.

Declarations

Funding. No external funding.

Conflict of Interest. None declared.

Ethics Approval. Not applicable. All datasets are publicly available benchmarks.

Data Availability. IIIT-D [17], CUHK/CUFS [18], CUFSF [19], and AR [20] are publicly available. Code released upon acceptance.

Author Contributions. D.U.N. conceived the DAAFSS framework, designed the architecture, implemented all components, conducted all experiments, and wrote the manuscript. M.A. contributed to the literature review, experimental design, result analysis, and manuscript revision. Both authors read and approved the final manuscript.

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